

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester III Examination, 2023-24

Paper No. CHB-301: Organic & Physical Chemistry-II

Time: 3 Hours

Maximum Marks: 70

Instructions: Answer five questions in total, including Question 1 from each section which is compulsory. Use separate answer books for Section A and Section B.

SECTION A

(Organic Chemistry-II – Marks: 35)

Question 1. Answer all parts of the following:

[5 × 2 = 10]

- (a) Classify the following species as aromatic, antiaromatic, or nonaromatic: (i) Cyclopropenyl anion, (ii) Cyclopentadienyl cation, (iii) Tropylium ion, and (iv) 1,3,5,7-cyclooctatetraene.
- (b) Identify structures A and B along with the mechanism of their formation:
- $$\text{Ethylene oxide} \xrightarrow{\text{CH}_3\text{MgBr, THF}} A \xrightarrow{\text{H}_2\text{O}} B$$
- (c) Arrange the following compounds in decreasing order of acid strength, with suitable reasons: phenol, *p*-nitrophenol, *p*-cresol, and *m*-nitrophenol.
- (d) Draw the molecular orbital picture of benzene, indicating all nodal planes.
- (e) What is DDT? Write down the reaction mechanism of its preparation from chloral and chlorobenzene.

Question 2.

- (a) Discuss the following reactions along with mechanisms:

[3 + 3 = 6]

- (i) Beckmann rearrangement
- (ii) Baeyer-Villiger oxidation

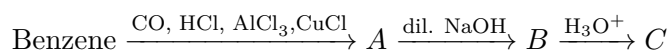
- (b) Provide a suitable mechanism for the following reaction:

[3]



- (c) Identify the products (A–G) in the following reaction sequence:

[3]

**Question 3.**

- (a) Explain why acetanilide undergoes mainly *p*-electrophilic substitution reactions, while aniline in strongly acidic medium exclusively produces *m*-products. [6]
- (b) Describe the synthesis of Novolac and Bakelite from phenol, including reactions and mechanisms. [6]
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Question 4.

- (a) An organic compound (A) with molecular formula C_7H_8O reacts with neutral $FeCl_3$ to give a violet color. A reacts with CO_2 and $NaOH$ at high pressure to give B, which on acidification yields C. Identify A, B, and C, and write the mechanisms involved. [6]
- (b) Identify the products from the following reactions and discuss the mechanism of their formation: ... [6]
- (i) Pyridine $\xrightarrow{NaNH_2, 100^\circ C}$?
- (ii) Pyrrole $\xrightarrow{CHCl_3, KOH}$?

SECTION B

(Physical Chemistry-II – Marks: 35)

Question 5. Answer all parts of the following: [5 × 2 = 10]

- (a) Construct a cell from which one can measure the standard reduction potential of the $\text{Ag}^+(\text{aq}) + \text{e}^- \rightarrow \text{Ag}(\text{s})$ half-cell reaction.
- (b) Transference numbers of ions depend on concentration and temperature. Explain.
- (c) The equivalent conductivity of CH_3COONa , HCl , and NaCl at infinite dilution at 25°C are given as 91.0, 426.2, and $126.5 \text{ ohm}^{-1} \text{ cm}^2 \text{ equiv}^{-1}$, respectively. Calculate the equivalent conductivity of acetic acid at infinite dilution.
- (d) Draw with brief explanation the approximate curve for the conductometric titration of NH_4OH solution with HBr solution as titrant.
- (e) Under what conditions does the Gibbs phase rule become $F = C - P + 1$?

Question 6.

- (a) Show that for a solution of a single electrolyte, the mean ionic activity ($a_{\pm}$) is related to its molality (m) by $a_{\pm} = \gamma_{\pm} m_{\pm}$. Calculate $a_{\pm}$ for a 0.01 mol/kg solution of $\text{Al}_2(\text{SO}_4)_3$. [6]
- (b) Describe the construction and working of the Quinhydrone electrode. What are its advantages and limitations? [6]

Question 7.

- (a) The e.m.f. of a cell consisting of a hydrogen electrode and SCE was found to be 0.682 V at 25°C in a buffer solution. The e.m.f. increased to 0.794 V when another buffer solution was used. What was the pH of the second buffer? [6]
- (b) How do buffer solutions resist changes in pH upon addition of small amounts of acid or base? Derive the Henderson-Hasselbalch equation for an acidic buffer. [6]

Question 8.

- (a) Describe the procedure to determine the equilibrium constant of the reaction $\text{KI} + \text{I}_2 \rightleftharpoons \text{KI}_3$ in aqueous medium using the Nernst distribution law. [6]
- (b) Derive Bragg's equation for X-ray diffraction. If the first-order diffraction from a crystal plane occurs at a glance angle of 30° using Cu-K_{α} radiation ($\lambda = 1.54 \text{ \AA}$), calculate the interplanar spacing (d). [6]

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